Each project is assigned a relative difficulty number (2-5), 5 being “most challenging”.

1. Comparing Bootstrap vs. Asymptotic Confidence Intervals (DN=3)

For sample means and medians from skewed and symmetric distributions:

- Simulate repeated samples and compare coverage probabilities of t-based CIs vs. percentile/bootstrap-t CIs.
- Discuss performance as sample size increases and tail heaviness varies.

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2. Random Walks and the Law of Large Numbers (DN=3)

Simulate 1D and 2D random walks:

- Examine convergence to zero drift under various step distributions.
- Estimate hitting times and visualize path variability.
- Extend to biased walks or absorbing barriers.

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3. Resampling Methods for Model Validation (DN=4)

Compare **k-fold cross-validation**, **bootstrap**, and **LOOCV** for regression model performance estimation.

- Use synthetic data where the true model is known.
- Examine bias and variance of each estimator of prediction error.

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4. Bayesian Simulation and Approximation (DN=4)

Use **Monte Carlo sampling** or **Metropolis–Hastings** to approximate a posterior distribution for a parameter (e.g., binomial proportion, normal mean with unknown variance).

- Compare your simulated posterior to the analytical result (if available).
- Summarize posterior mean, credible intervals, and diagnostics.

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5. The Central Limit Theorem in Action (DN=2)

Simulate sums of random variables from several distributions (uniform, exponential, heavy-tailed).

- Demonstrate convergence toward normality using histograms and QQ-plots.
- Quantify convergence rate (Berry–Esseen bounds, empirical skew/kurtosis over n).

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6. Comparing Parametric vs. Nonparametric Density Estimation (DN=4)

Simulate data from known distributions and compare:

- Parametric (MLE under correct/incorrect assumptions)
- Nonparametric (kernel density, histograms)
  Use integrated squared error to evaluate performance and visualize fits.

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7. Stochastic Optimization via Simulation (DN=5)

Use simulation to solve an optimization problem under uncertainty — e.g., inventory management, staffing, or investment allocation.

- Compare deterministic vs. stochastic solutions.
- Implement a Monte Carlo-based gradient-free optimizer (like simulated annealing).